

ORIGINAL ARTICLE

Personal networks and psychological attributes: Exploring individual differences in personality and sense of community and their relationship to the structure of personal networks

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Action Editor: Stanley Wasserman

Abstract

The paper explores the relationship between personality characteristics, sense of community and the structure of personal networks. We collected personal network data from 100 adults, consisting of the information about their 45 alters and ties among them. In addition to the typically used bivariate correlations to analyze the relationship between individual psychological differences and network structure, we propose first the use of typologies of networks and personality and second modified versions of the triadic census in ego-networks. The results show that different personality types tend to occupy a different kind of networks and that newly introduced triadic measures show a relatively higher association with examined psychological attributes than global network measures. Overall, the Psychological Sense of Community showed higher associations with network measures than Big Five personality traits. The former was positively correlated with measures of closed triads among alters, but also with alters embedded in triads with weak ties. Regarding personality traits, Emotional Stability was positively correlated with strong closed triads and with the overall indicator of density. The elaboration of typologies and the census of triads are shown to be effective strategies for the description of personal networks, as well as for the analysis of individual psychological differences.

Keywords: personality traits, Big Five, personal networks, typology, triads, individual differences, psychological predispositions

1. Introduction

People utilize different interactive strategies that determine both their position in the social network and the structural properties of their personal networks (Krause et al., 2010). This notion has recently started to be investigated through the study of individual differences in social networks (Selden & Goodie, 2018). This is an innovative approach, in an area in which the emphasis on constraints that the social structure imposes on opportunities for interaction and, consequently, on individual behavior has traditionally predominated (Wellman, 1983).

Previous research has looked into the variability of the structure of personal networks in terms of socio-demographic characteristics, such as age, gender, socioeconomic status, and educational level (Roberts et al., 2008). The changes that take place throughout the life cycle have also

been examined, based on personal transitions and life events (Bidart & Lavenu, 2005; Dickens & Perlman, 1981; Ferrand, 1989; Morgan et al., 1997). The focus on individual differences in personality is recent. It is based on previous evidence of the influence of the psychological profile on the levels and styles of sociability (Digman, 1990; Furukawa et al., 1998; Roberts et al., 2008; Russell et al., 1997).

The approach that has been predominately used to explore the individual variability in the structure of personal networks is the model of the five major personality factors (Goldberg, 1993; Tupes & Christal, 1961, 1992). In this study, we rely on this background to explore the relationship between psychological attributes and the structure of personal networks. However, we aim to contribute to this body of research, first by including psychological attributes that theoretically may be more directly related with the network structure (psychological sense of community). Second, by using different analytical approaches that take advantage of typologies of both personality traits and network, and third, we additionally make an extension of previously used measures of ego-networks based on motif analysis (Kalish & Robins, 2006).

The results of our study suggest that psychological sense of community is associated with different aspects of the structure of personal network, while degree of association is similar or slightly higher than that of the five personality factors. Typological approach shows that individuals with different personality types have tendency to occupy different types of networks, while the introduced modifications of triadic measures give a more nuanced picture of the interplay between individual psychological differences and network structure.

2. The influence of the Big Five personality traits on the structure of personal networks

2.1 Extraversion

Extraversion is characterized by sociability and seeking the company of others. Extraverted people tend to be described as sociable, talkative, friendly and predisposed to experience positive emotions (McCrae & Costa, 1987). This makes them develop better social skills (Doeven-Eggens et al., 2008; Roberts et al., 2008), which help them both to start new relationships (Asendorpf & Wilpers, 1998; Roberts et al., 2008; Wrzus et al., 2017; Zhu et al., 2013), and to maintain them over time (Zhu et al., 2013).

Extraverted people gain reinforcement in interpersonal situations, which predisposes them to make friendly relationships (Feiler & Kleinbaum, 2015; Roberts et al., 2008; Selfhout et al., 2010), to actively seek opportunities for interaction with others (Asendorpf & Wilpers, 1998; Doeven-Eggens et al., 2008; Landis, 2016; Totterdell et al., 2008; Wagner et al., 2014; Wrzus et al., 2017; Zhu et al., 2013) and to make sure that their personal contacts know each other (Kalish & Robins, 2006).

This tendency towards sociability tends to be reflected in a more extensive and varied support networks (Burt et al., 1998; Cohen et al., 2000; Swickert et al., 2002; Wagner et al., 2014), with a higher frequency of interaction (Swickert et al., 2002) and a higher proportion of new contacts (Zhu et al., 2013). The studies with homogeneous samples have generally found a positive correlation between extraversion and the size of the personal network (Roberts et al., 2008; Selden & Goodie, 2018). However, extroverts' networks are not denser than those of the introverts, suggesting that they are not necessarily fostering ties among their acquaintances (Kalish & Robins, 2006).

2.2 Agreeableness

People who score high on this trait are characterized by a tendency to cooperate, based on empathy and altruism. Agreeableness normally corresponds to people described as trusting, sympathetic and considerate to others (McCrae & Costa, 1987). They are usually helpful, and seek the good

of the community, so they inspire confidence when requesting information or other types of resources (Liu & Ipe, 2010).

Greater agreeableness is related to the development of long-term positive affective relationships (Doeven-Eggens et al., 2008; Zhu et al., 2013), reducing both the likelihood and the intensity of interpersonal conflicts (Asendorpf & Wilpers, 1998; Wrzus et al., 2017; Zhu et al., 2013), and improving the perceived quality of friendship (Demir & Weitekamp, 2007). People who excel in this personality factor attract friendships and have a greater chance of being chosen by others to start a relationship (Selfhout et al., 2010; Zhu et al., 2013).

In the studies with individuals who experience a personal transition process, a correlation has been found between agreeableness and the size of the egocentric network, mainly due to new non-kin contacts (Wagner et al., 2014). However, there is not yet much evidence in this field.

2.3 Conscientiousness

Conscientiousness refers to a general tendency towards self-control and discipline. People who score high on this trait are conscientious, perseverant, reliable and disciplined at work (McCrae & Costa, 1987). They are characterized by greater motivation for achievement and focus on the resolution of tasks (Fang et al., 2015; Zhu et al., 2013).

It has been suggested that conscientiousness is reflected in more stable and lasting relationships, as well as in a lower proportion of new contacts (Asendorpf & Wilpers, 1998; Baker & McNulty, 2011; Zhu et al., 2013). Indeed, several studies show that there is a significant relation between responsibility and listing more family members in the personal network and with a greater stability of these members over time (Doeven-Eggens et al., 2008; Wagner et al., 2014).

2.4 Emotional stability (neuroticism reversed)

At the negative extreme of emotional stability is neuroticism. Neuroticism is characterized by a tendency to experience negative emotions, along with a greater vulnerability to stress and depression. Among other aspects, it usually entails emotional insecurity, distrust, constant worry, anxiety and other forms of affective instability (McCrae & Costa, 1987). People who score high on this trait tend to evaluate their experiences negatively (Doeven-Eggens et al., 2008), express more negative emotions (Fang et al., 2015), and are less assertive in their interactions (Roberts et al., 2008).

Emotional instability is usually associated with insecure, hostile, and lower quality relationships (Wrzus et al., 2017). In fact, people with high neuroticism are commonly perceived as “high cost interaction partners” (Fang et al., 2015, p. 1245) and are less attractive as friends. As a result, they have less opportunities to develop the skills of initiating and maintaining relationships, while also experiencing more interpersonal conflicts (Demir & Weitekamp, 2007).

Although neurotic people declare themselves less predisposed to form new relationships, there is no evidence that shows a significant impact on the size and composition of personal networks (Selden & Goodie, 2018). Those who score high in neuroticism seem to be less predisposed to the connection between members of diverse groups in their personal network (Kalish, 2008), while they perceive less transitivity and greater disconnection in networks of strong ties (Kalish & Robins, 2006).

2.5 Openness to experience

High scores on this trait are an indicator of a personal inclination for change, creativity and imagination. Openness to experience is usually represented through adjectives such as original, imaginative, curious and preference for variety (McCrae & Costa, 1987). These are individuals who adapt to change easily and who show tolerance for novelty.

Individuals who are open to new interpersonal experiences may also be more available to establish new contacts, explore new contexts of interaction and have less consolidated relationships

Table 1. Characteristics of personal networks and position in complete networks according to the Big Five personality traits.

Personality traits	Personal networks	Socio-centric networks
Extraversion	They have larger friendship networks	Occupy positions of greater centrality in friendship, advice and adversarial networks
	They have a higher proportion of new contacts	Have higher indicators of out-degree (activity)
Agreeableness	They have larger non-kin networks	Occupy positions of greater centrality in friendship networks
		Obtain higher indicators of in-degree (popularity)
		Tend to connect different groups
Conscientiousness	More family members in the personal network, more stable over time	Can be key actors in advice and instrumental networks
Emotional stability (neuroticism reversed)	It does not seem to affect the size or composition of the personal network, but rather the perception of it	In some cases, it is related to a greater centrality in advice networks
	Highly neurotic people perceive less connectivity in their personal network	Obtain higher in-degree indicators in adversarial networks
Openness to experience	It is related to the incorporation of new contacts in moments of personal transition	Obtain higher in-degree indicators in adversarial networks

over time (Wagner et al., 2014). This could put them in a position to act as intermediaries, when they come into contact with new social environments where other individuals are not connected.

It has been observed that people with greater openness to experience have a higher proportion of new contacts (Zhu et al., 2013), establish ties with people from different contexts (Fang et al., 2015; Landis, 2016), maintain support relationships for less time, and tend to have a lower frequency of contacts (Wrzus et al., 2017). Although not direct effect has been noted on the size of the network, there is a positive relation between the level of openness and the establishing of new relationships at times of personal transition (Selden & Goodie, 2018; Wagner et al., 2014).

3. The influence of the Big Five personality traits on the centrality of individuals in socio-centric networks

To complete our examination of the Big Five model, below we review the studies that have used socio-centric designs, or complete networks. Although the focus of our research is personal networks, these other studies can also reveal the relational dynamics of interest based on personality traits. In Table 1, we summarized some of the most consistent evidences both in personal network surveys and in the analysis of complete networks, organized according to the five basic personality traits.

3.1 Extraversion

Utilizing data from socio-centric networks, various studies have documented that extroverts tend to have a greater centrality in friendship networks (Feiler & Kleinbaum, 2015), advice networks (Klein et al., 2004; Regts & Molleman, 2016), and adversarial networks (Klein et al., 2004). Extroverts are especially active in networks and generally obtain higher scores in out-degree centrality (Selden & Goodie, 2018).

3.2 Agreeableness

In the case of people with high scores in agreeableness, it has only been found that they are more central in friendship networks (Klein et al., 2004). Although the evidence is still scarce, in some

cases, they also show a greater probability of being chosen by others as friends (Selfhout et al., 2010), or as relational leaders (Emery, 2012; Emery et al., 2013), obtaining a high in-degree. This coincides with another finding suggesting that they also act as connectors between groups that are separated from each other (Battistoni & Fronzetti Colladon, 2014).

3.3 Conscientiousness

Conscientiousness is often a good indicator to detect individuals who play a key role in instrumental networks (which usually involve giving timely advice, transmitting quality information, or solving tasks) (Battistoni & Fronzetti Colladon, 2014; Emery, 2012; Emery et al., 2013).

3.4 Emotional stability (neuroticism reversed)

People with less emotional stability (or with more neuroticism) are less likely to facilitate the development of relationships between people or groups that are disconnected from each other (Battistoni & Fronzetti Colladon, 2014). In some cases, a high score in neuroticism is related to a greater centrality in the instrumental networks and advice networks in work contexts. This has led to suggest that, under certain circumstances, it may be beneficial in organizational contexts (Selden & Goodie, 2018). People scoring high in neuroticism may also have greater in-degree centrality in adversarial networks (Klein et al., 2004; Xia et al., 2009).

3.5 Openness to experience

High scores in openness seem to predict a good incorporation in distributed and long-distance communication networks, in which it is important to relate disconnected individuals to each other (Xia et al., 2009). However, it has been observed that this trait has a negative relationship with centrality in friendship networks, and positive in adversarial networks (Klein et al., 2004).

4. Identifying the need for a different analytical approach

Research findings concur that some psychological attributes are associated with network variables. Nonetheless, the pattern of relationship found varies. This could be a result of the different methodologies employed for constructing personal networks and small homogenous samples (e.g. students, employees). The strength of correlation rarely exceeds 0.3, and recent meta-analysis of socio-centric studies (Fang et al., 2015) show that, although significant, the shared variance between psychological attribute variables and social networks variables is modest—rarely exceeding 5%. Despite being small in absolute terms, this finding demonstrates that the structure of our network is 1/20 related to our psychological attributes, which is not to be considered trivial given that our network is made of other people with their own psychological attributes. However, the research done so far is not addressing all psychological attributes and their influence on the individual's overall network structure. The findings of previous research refer only to specific dimensions of personality and their association with specific structural characteristics of an ego-network, without taking the other dimensions simultaneously into account. For example, the majority of research on extraversion and network size show that the two are positively correlated, which is highly expected given the definition of extraversion. On the other hand, an extrovert who is also emotionally unstable will have different proclivity, opportunity, and success in developing and maintaining new ties than an extrovert who is also emotionally stable. In other words, in most social interactions, we manifest several psychological attributes simultaneously (e.g. extraversion, emotional stability, and agreeableness). That is why research should look at the effect of psychological attributes on social networks in a way that considers the combination of traits. Although the Big Five personality traits are conceptualized as independent, existing research has shown that

considerable correlations among the traits exist (van der Linden et al., 2010). Although the majority of studies have been based on the broad “Big Five” traits, in some studies, narrower trait-like constructs from social psychology, that capture values and beliefs of an individual, have also been used, e.g. locus of control, self-monitoring, self-efficacy (Kalish & Robins, 2006). Thus, when making assumptions about the relationship between psychological attributes and social networks, we should also be aware that previous research has not examined all the psychological attributes that may be significantly related with social networks.

5. This study

In this study, we explore how personality traits relate to the structural properties of personal networks. For this, we rely on the findings of previous research, reviewed in the previous sections, linking singular personality variables with the specific indicators of centrality, size, homophily and cohesion of personal networks (Roberts et al., 2008; Selden & Goodie, 2018). The novelty of our approach is summarized in three points: in addition to the Big Five personality traits, we investigate whether there is a relationship between network structure and Psychological Sense of Community, using a typological approach in analyzing the relation between personality traits and network structures, and finally, using a motif analysis of ego-networks to shed more light on the relationship between psychological attributes and personal network structure.

5.1 Investigating psychological sense of community and its relation to personal network structure

Psychological sense of community refers to the feeling of belonging to a collective, in which the needs of the members are expected to be met through a cooperative commitment (Sarason, 1974). Together with a sense of belonging, the members of the community develop a shared emotional connection, their needs are addressed and they are able to influence the whole group (McMillan & Chavis, 1986).

Social cohesion and the sense of belonging have been connected conceptually with the density of the social network (Maya-Jariego, 2004). However, this association has hardly been researched empirically with structural properties indicators, be it of complete networks or personal networks. The existence of a moderate correlation has been proven in some cases (Maya-Jariego & Holgado, 2015). It is expected that other indicators of structural cohesion are also related to sense of community. In this study, we carry out a systematic exploration in this respect, which allows for the comparison of the relationship between network indicators and sense of community with the relationship between network indicators and other psychological attributes that were researched previously to a greater extent (e.g. personality traits).

5.2 Typological approach

Structural network properties analyzed in previous research are strongly associated with each other. In this respect, comparison of networks through isolated characteristics is an empirical approach with important limitations. For example, an outgoing person may tend to present their contacts to each other, which increases the density of his network. However, he is also more likely to make new contacts, which not only increases the size of the personal network, but also indirectly, the number of relationships necessary for the network to also increase in density. For this reason, analytic strategies have been proposed to address the configuration of networks as a whole, such as typologies based on cluster analysis (Bidart et al., 2018; Giannella & Fischer, 2016; Maya-Jariego, 2002) and triad census (Kalish & Robins, 2006). The factorial analysis of the structural properties of personal networks has differentiated the cohesion, fragmentation and integration of personal networks as significant dimensions of variability (Lozares et al., 2013; Maya-Jariego & Holgado, 2015).

5.3 Triadic measures

The other approach, to circumvent the problem of describing a network with many different structural measures, is to focus on the network's smallest possible units, triads. The term triad is often associated with the work of Simmel (1950) who stated that a group of three people is a basic unit of network analysis. Siltaloppi & Vargo (2017) in an overview of the triad concept state that triads cannot be defined merely as systems of three actors, but at least, by the coexistence of two ties between three associated actors. These authors suggested that triadic analysis is not limited to the specific systems of exactly three actors but applicable to any system of at least three actors. Consequently, triads can be used for analyzing multiple forms of triadic relationships in systems like ego-networks.

This kind of mesoscopic network approach could be fruitful for shedding some light on the psychological attribute—social networks connection, as triads are easily adapted to measuring and theorizing (Vinacke & Arkoff, 1957). Kalish & Robins, (2006)¹ introduced triadic measures (in the following text addressed as KR triads) derived from ego-networks that are supposed to capture differences between networks which are not captured by other measures (e.g. size, density, efficiency) and provide more accurate and informative descriptions than global network measures. The authors found that this measure could be related with individual differences in personality traits. The measure is based on proportions of several possible configuration of triads with strong and weak ties in ego-network (including ego). One of their findings was that higher proportion of strong closed triads is positively associated with extraversion and negatively with neuroticism. The opposite holds for weak closed triads. Also, individual and group focus was positively related with proportion of open and closed triads, respectively.

In contrast to the original KR triadic measure, which looked at triads in an ego-network that included ego, in this paper we are presenting three variants of KR measure that take into account triads in ego-networks, but without ego. Also, the original KR measures were developed and used in networks with varying numbers of alters (e.g. Staiano et al., 2012), while we have developed three versions of this measure that are also applicable to networks with a fixed size (described in the Methodological appendix in the Supplementary Material).

When an alter i is connected to two other alters, j and k , the triad that describes the relationship between these three alters is denoted by a three-letter combination. As a notation system, S represents strong ties and W represents weak ties. A total of seven possible triads can occur between alters in egocentric networks, four closed triads: SSS, WWW, SSW, and SWW; and three open triads: WWN, SSN, and SWN, where N stands for an absence of tie between alters (shown in Figure 1). Differently from KR triads, here the ordering of letters is not informative, for example, triads SSW, WSS and SWS are equivalent as all letters indicate the strength of tie between any pair of alters. Seven distinct triads fall under one of the two groups: closed or open triads which represent a purely structural description without taking into account the strength of present ties in the triad.

We propose three different ways of calculating the triadic census for alters in ego-network with fixed size: KR variant I, KR variant II, and KR variant III. The procedure of calculating these measures is described in detail in the Methodological appendix in the Supplementary Material and is done with open and free software Python.

5.4 Research questions

In this research, we combine the construction of typologies and the census of triads to describe the relationship of personality traits and psychological sense of communities with the structure of personal networks. Specifically, we aim to answer the following questions:

- (1) What is the type and the strength of an association between the variability in cohesion, fragmentation and integration of personal networks and individual differences in the Big Five personality factors?

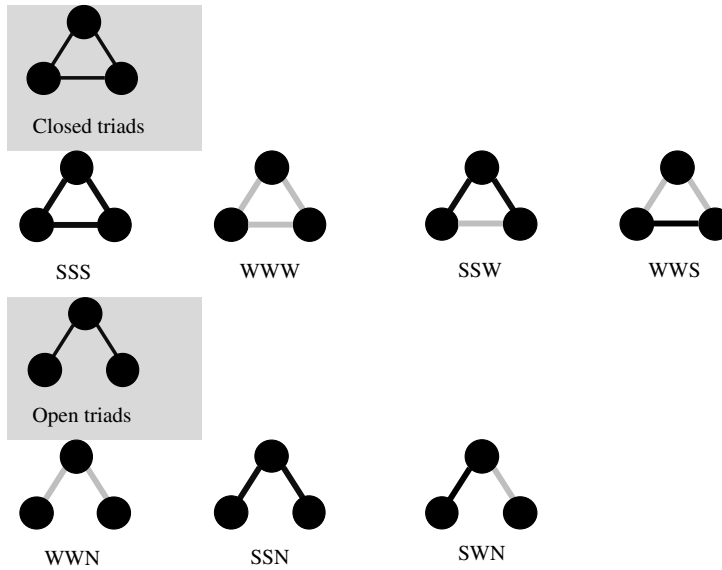


Figure 1. Two groups of seven possible triads that can occur in egocentric network among ego's alters when considering the strength of ties. The triads on grey background present two distinct group of configurations, open and closed. In the triads with no background, a black line represents a strong tie, a grey line represents a weak tie, while all nodes are an ego's alters. The three letters are unique code for each triad (S, strong; W, weak; N, no tie is present).

- (2) How is the density and cohesion of personal networks related to Psychological Sense of Community?
- (3) What is the type and the strength of an association between the presence of some triadic configurations (e.g. closed triads) in personal networks and both personality traits and Psychological Sense of Community?
- (4) Do typologies and census of triads effective strategies provide new insights about individual differences in personal networks?

6. Method

6.1 Participants

We surveyed 100 adults living in the metropolitan area of Seville, of which 60 were women and 40 men, with an average age of 38.74 years ($SD = 14.64$). Socio-demographic differences were observed, both in terms of education: 46 interviewees had completed a university degree, 46 secondary school, and 8 primary school; and employment: almost two-thirds (64) were employed, 24 were students, and 14 were inactive (unemployed or retired). The data was obtained through face-to-face interviews lasting approximately 1 hour. The respondents participated on a voluntary basis and were notified that the data would be treated in an aggregated manner, guaranteeing the confidentiality of the information.

In the surveys using the five personality factors model, samples of university students predominate, although previous evidence demonstrates that the variability of personal networks can be influenced by demographic aspects (Roberts et al., 2008). For this reason, in this study, we seek to introduce some diversity in the profile of the respondents.

6.2 Instruments

The respondents answered two psychometric scales, a personal networks interview and a list of questions on socio-demographic aspects. Next, we describe each of these instruments.

6.2.1 Ten-Item Personality Measure (TIPI) (Gosling et al., 2003)

This 10-item scale measures the Big Five personality dimensions: neuroticism, extraversion, openness to experiences, agreeableness, and conscientiousness (N, E, O, A, and C, respectively). It is a measure with two items referring to a personality trait for each of the five dimensions (e.g. “extraverted, enthusiastic,” “sympathetic, warm,” “calm, emotionally stable”) with two adjectives. The participants score each item on a scale ranging from 1 (“Strongly disagree”) to 7 (“Strongly agree”). Each personality dimension is evaluated with two items. The overall score of the scale is the average of the response of each participant of the 10 items.

The shortness of the TIPI means no high reliability coefficients are obtained. However, the instrument obtains good indicators of convergent validity, discriminant validity and test–retest reliability. The scale allows an efficient evaluation of personality traits (Gosling et al., 2003). The Spanish version has been validated with good results (Renau et al., 2013). In general, the Big Five model shows a great consistency in different cultural contexts (Benet-Martinez & John, 1998).

6.2.2 Psychological Sense of Community Scale (PSC) (Jason et al., 2015)

This scale evaluates psychological sense of community through three factors: Entity, Membership, and Self. This is an inventory of nine items that respondents score on a Likert-type scale from 1 (“Strongly disagree”) to 6 (“Strongly agree”). In our case, it was applied referring to the neighborhood where the respondent resided at the moment of the interview. The items allow to assess the characteristics of the community (Entity) (e.g. “this neighborhood is a good neighborhood”); the relations between the members of the collective (Membership) (e.g. “the neighbors can get help from other neighbors if they need it”); and the emotional connection with the group (Self) (e.g. “this neighborhood is important to me”). The overall score of the scale is the average of the response for the nine items. The original validation yielded a reliability alpha coefficient of 0.923, whereas in this study we obtained a coefficient of 0.871.

6.2.3 Network measurement

To evaluate the personal network, the following name generator was used: *Please give me a list of 45 people with whom you have a relationship throughout the week. I am interested in those people with whom you have frequent and habitual contact. They can be colleagues, neighbors, relatives, friends, or people with whom you share hobbies. They can be from your neighborhood, from nearby neighborhoods, or even from more distant ones. It is important that they are the 45 people with whom you have a more frequent relationship.* Then, for each pair of alters, the respondents were asked to rate the type of relationship on four levels of intensity, where 0 corresponds to “they do not know each other”; 1, “they know each other”; 2, “they have a relationship”; and 3, “they are friends or have a strong relationship.” Therefore, we generated 100 symmetric and valued matrices of alter–alter relationships, with 45 actors each.

The establishment of a fixed number of alters introduces a bias in the structural properties of networks (Neal & Neal, 2017). However, it is a standardization method that facilitates the processing of data and has been shown as a reliable method of socio-metric nomination (McCarty, 2002; Molina et al., 2014). It has been used with all types of populations for more than a decade and a half, with good results of validity and reliability. In addition, it is very advantageous for the standardization and comparability of personal networks (Maya-Jariego, 2018).

6.2.4 Socio-demographic variables

Finally, the participants provided information on their professional situation, educational level, people with whom they live, type of housing (owned or rented), and the time they have lived in the neighborhood of their current residence. In addition to the data about gender and age, each respondent was asked to indicate if they collaborate with a community organization, as well as, where appropriate, the time and format of participation (online or in person).

Table 2. Descriptive statistics for personality and sense of community measures.

	Minimum	Maximum	Mean	Standard deviation	Skewness	Kurtosis
<i>Ten-Item Personality Measure (TIPI-10)</i>						
Extraversion	1	7	4.84	1.47	−0.284	−0.659
Agreeableness	2.5	7	5.61	1.04	−0.703	0.155
Conscientiousness	3	7	5.60	1.15	−0.479	−0.874
Emotional stability	1	7	4.93	1.43	−0.626	−0.421
Openness	1	7	5.06	1.34	−0.661	0.263
Mean TIPI-10	3.6	7	5.21	0.67	0.094	0.030
<i>Psychological Sense of Community Scale (PSC)</i>						
Entity	1	6	4.73	1.23	−0.969	0.247
Membership	1.33	6	4.42	1.09	−0.659	0.147
Self	1	6	4.70	1.24	−1.009	0.355
Mean PSC	2	6	4.61	0.98	−0.707	0.332

Note. Emotional stability is the positive dimension of the factor “neuroticism.”

6.3 Procedure and data analysis

The 100 matrices of personal networks were analyzed with UCINET 6.627 (Borgatti et al., 2002). To summarize the structural properties of each personal network, four specific indicators were calculated: density, degree centralization, number of components, and number of cliques. Normally the indicators of centrality, cohesion and groups are highly correlated with each other. This is why some previous studies have tried to determine which are the fundamental dimensions of variability. In two community surveys with different representative samples, it was found that density/centralization, cliques, and components report different factors (Lozares et al., 2013; Maya-Jariego & Holgado, 2015), which correspond to cohesion, integration and fragmentation of the network, respectively.

Next, we created a database in which, together with the response to the psychometric scales and the socio-demographic variables, the four summary indicators of each respondent's personal network were included.

We used *quick cluster* analysis to develop typologies. In our approach to cluster analysis, we used the information from the literature, our data, and combined it with the methodological considerations and other analyses done in this study. In this way, the classification “is partly constructed and partly discovered” (Maya-Jariego, 2002, p. 4). The criteria variables were selected considering previous research, descriptive analysis, while opting to choose the variables which are not highly correlated with each other (below 0.7). Using completely uncorrelated variables for cluster analysis (e.g. density and cliques), the best quality solution with the highest Silhouette score led to a high number of clusters (seven) for our sample size. Therefore, we used transitivity instead of density. Transitivity is highly related with density ($r_s = 0.65$) because it also captures the network connectivity, but unlike density it considers its triadic level rather than its dyadic level, which is more in line with our other analytical strategy which is focused on triads.

7. Results and discussion

7.1 Descriptive data of individual differences

Table 2 summarizes the descriptive statistics of personality and sense of community. The data present a slightly asymmetric distribution, although very concentrated around the average scores. In both cases, there is a positive bias in the assessment of personality traits and sense of belonging.

The five personality traits are generally independent of each other. The statistically significant correlations observed are between extraversion and openness to new experiences ($r = 0.397$,

Table 3. Pearson correlations between psychological attributes and personal network measures in the network of acquaintances (NA) and the network of strong ties (ST).

	Density		Degree centralization		Number of components		Number of cliques	
	NA	ST	NA	ST	NA	ST	SA	ST
<i>Personality</i>								
Extraversion	0.105	−0.027	−0.041	0.001	−0.016	0.039	0.065	−0.020
Agreeableness	0.103	−0.010	−0.271**	−0.006	−0.023	0.087	−0.155	0.037
Conscientiousness	0.072	−0.092	−0.004	−0.125	−0.034	0.133	−0.130	−0.108
Emotional stability	0.211*	0.167	−0.193	0.019	0.062	0.036	−0.169	−0.087
Openness	0.046	0.084	−0.043	−0.066	−0.064	−0.121	−0.006	0.184
Mean TIPI-10	0.209*	0.058	−0.201*	−0.062	−0.024	0.056	−0.137	0.002
<i>Sense of community</i>								
Entity	0.237*	0.166	−0.027	0.152	−0.132	−0.070	−0.052	0.010
Membership	0.265**	0.148	−0.112	0.071	−0.44	0.017	−0.214*	−0.042
Self	0.243*	0.119	−0.60	0.093	−0.139	−0.028	0.100	−0.053
Mean PSC	0.299**	0.174	−0.078	0.129	−0.129	−0.035	−0.059	−0.034

Note. Emotional stability is the positive dimension of the factor “neuroticism.”

* $p < 0.05$.

** $p < 0.01$.

$p < 0.01$), and between agreeableness and emotional stability ($r = 0.531$, $p < 0.01$). On the other hand, three factors of PSC have a moderate correlation with each other (with $r = 0.417$, $p < 0.01$; $r = 0.499$, $p < 0.01$; and $r = 0.655$, $p < 0.01$).

7.2 Relationship of network indicators with psychological attributes

Table 3 presents the correlations between psychological attributes (personality traits and psychological sense of community) and the structural properties of personal networks. The overall score on the personality scale has a positive association with density ($r = 0.209$, $p < 0.05$) and a negative relation with degree centralization ($r = -0.201$, $p < 0.05$). That is, the personality profile shows a significant relationship with the indicators of cohesion, but not with the other two factors. More specifically, respondents with greater emotional stability have denser personal networks ($r = 0.211$, $p < 0.05$), while a higher score in agreeableness is significantly related to lower scores in degree centralization ($r = -0.271$, $p < 0.05$).

On the other hand, Psychological Sense of Community has a positive relationship with the density of the personal network ($r = 0.299$, $p < 0.01$). This is the strongest correlation observed, which is confirmed by the fact that all the factors of the PSC scale maintain a significant correlation with density. Finally, the Membership factor has an inverse relationship with the number of cliques ($r = -0.214$, $p < 0.05$).

In the context of ego-network without ego and its ties, density is equivalent to the proportion of closed triads with ego. Density is positively correlated with four out of five Sense of Community measures, and it has shown similar relationship with emotional stability, and the mean score of the TIPI10 scale (Mean TIPI10). The opposite pattern exists for the complementary measure of density of the missing ties. Transitivity showed a similar pattern of results as density, with correlation with the Big Five traits being slightly higher, and correlations with Sense of Community traits slightly lower than for density.

7.3 Relationship of triadic measures with psychological attributes

For the analysis of triads, we made three methodological variants: (1) triads as proportions of all possible triads, (2) triads as proportions of all existing triads in the ego-net, and (3) triads as

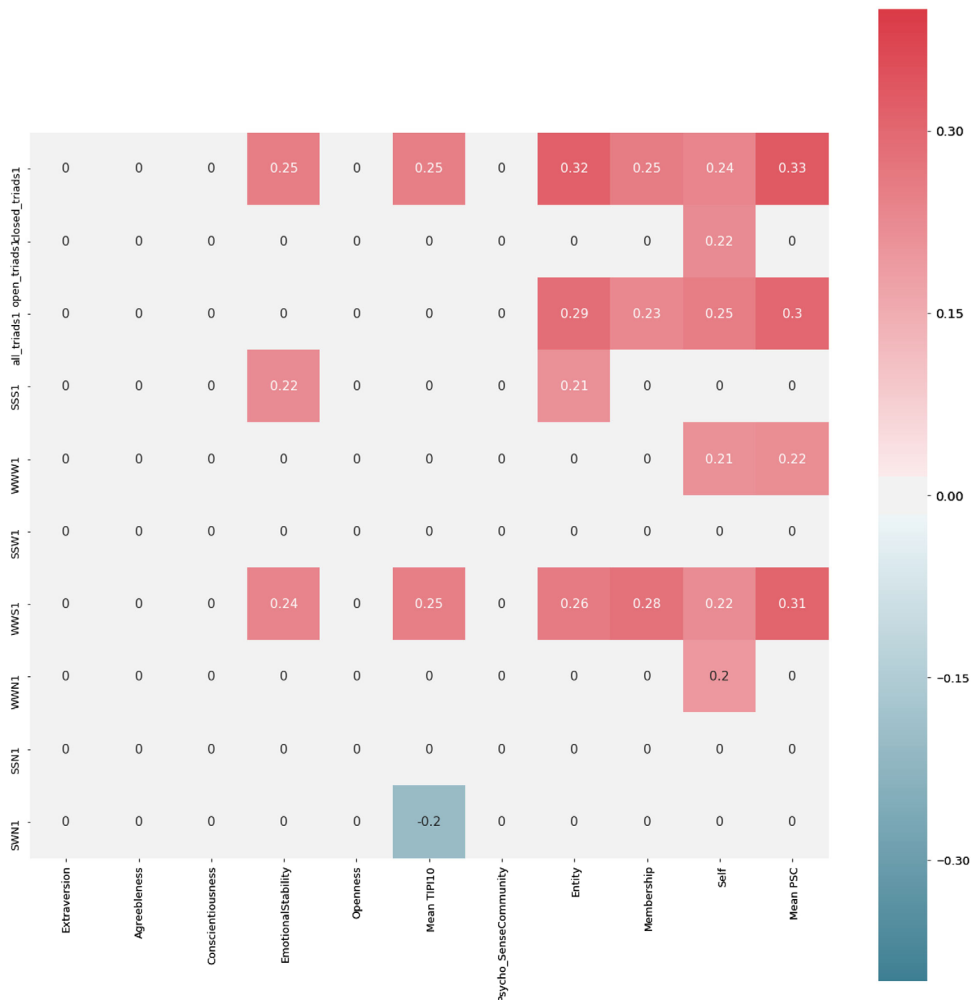


Figure 2. Significant correlations between KR variant I measures and psychological attributes (based on 1000 permutations).

z-scores in comparison to the individually tailored null model. The rationale and details of each approach are described in the Methodological appendix in the Supplementary Material. In the results presented in this section, all correlations are Spearman's rho coefficients, and significance testing is based on the 1000 permutations. If the value of coefficient had a percentile value higher than 97.5 or lower than 2.5, it was considered significant at the level of $p = < 0.05$. The strength of correlations is mostly low (below 0.4), as previous research would suggest and due to a small sample size, we were not able to detect correlations below 0.16.

The first variant showed the greatest number of significant associations, and of greater intensity as well. In Figure 2, KR triads of ego's alters, as proportions of all possible triads and their correlations with psychological attributes, are shown. The measures of the Sense of Community seem to be more connected with KR triadic measures than personality traits: closed triads are positively correlated with Entity, Membership, Self, and the mean of thePSC, while open triads are correlated to a similar degree only with Self. Strong closed triads (SSS) are correlated only with Entity, while weak closed triads (WWW) are related positively with both the mean score of the whole scale (Mean PSC) and Self. Additionally, weak open triads (WWN) are correlated with Self.

Table 4. Three types of personal networks: Centroids of the final conglomerates.

	<i>Cluster 1 (N = 16): Low transitivity, high number of cliques</i>	<i>Cluster 2 (N = 42): Low transitivity, average number of cliques</i>	<i>Cluster 3 (N = 42): High transitivity, small number of cliques</i>
Transitivity	0.69	0.67	0.84
Cliques	44.19	21.64	12.33

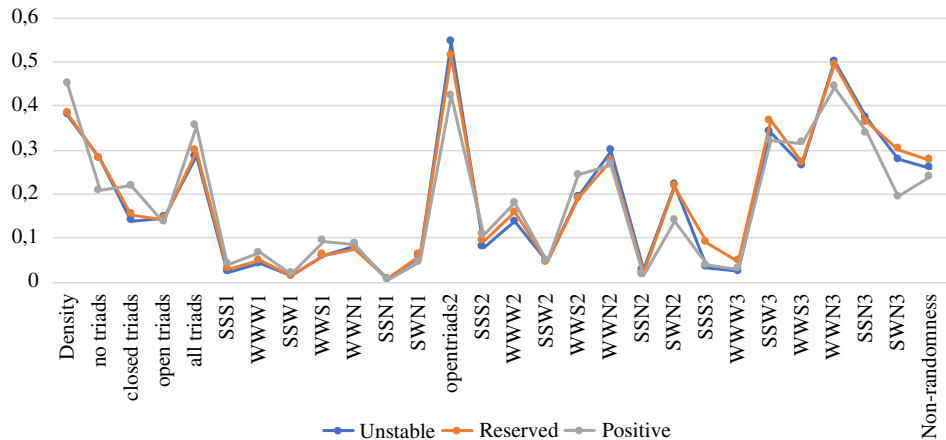


Figure 3. Differences in density and KR measures among three personality types.

These findings taken together suggest that, for the sense of community, alters embedded in triads with weak ties seem to be of importance.

The second variant showed more significant coefficients with personality factors than the first variant (see Figure 2 in the Methodological appendix in the Supplementary Material). This general pattern may imply that by controlling for triads present in the network, triadic measures tend to show relatively more associations with personality traits rather than with the components of sense of community. Specifically, the SWN (a “mixed” triad) seems to be the most relevant in this regard, as it is negatively correlated with E, C, emotional stability and the mean score of TIPI scale. In fact, the latter is the highest detected correlation between network and personality in this study.

In the third KR variant, in which we control for the density of the personal network and the individual tendency to assign strong or weak ties, the smallest number of significant associations is observed (see Figure 3 in the Methodological appendix in the Supplementary Material). Yet, it is interesting to note that Entity and the mean of the sense of community scale are positively correlated with WWW configurations, again suggesting that having weakly connected alters is an important aspect of psychological sense of community. This is a surprising finding due to lack of any connection of that group of psychological attributes with open or closed strong triads. This combination of results suggests that weak ties between alters are more important than strong ties between alters for some aspects of the subjective feeling of community (Entity) and the overall mean of sense of community scale.

7.4 Types of personal networks

We classified personal networks using transitivity and number of cliques as criteria variables. To do this, we applied *k*-means cluster analysis, with a maximum of 10 iterations. The solution of four categories showed good Silhouette score (0.426) indicating that a structure has been found and the clusters are not overlapping.

As shown in Table 4, the cluster 1 is the smallest and it describes networks with a high number of cliques and relatively lower transitivity. The second and third clusters are more than double in

Table 5. Three personality profiles: Centroids of the final conglomerates.

	Cluster 1 (<i>n</i> = 24)	Cluster 2 (<i>n</i> = 38)	Cluster 3 (<i>n</i> = 38)
	Unstable	Reserved	Positive profile
Extraversion	5.52	3.43	5.83
Agreeableness	4.50	5.89	6.04
Conscientiousness	5.46	5.59	5.71
Emotional stability	3.02	5.39	5.68
Openness experiences	5.02	4.29	5.86

Note. Emotional stability is the positive dimension (reverse scored) of the factor “Neuroticism.”

Table 6. The contingency table showing the relationship between personality types and network types with expected frequencies in the brackets.

Ego-network clusters/ personality clusters	Low transitivity, high number of cliques	Low transitivity, average number of cliques	High transitivity, small number of cliques
Unstable	7 (3.8)	12 (10.1)	5 (10.1)
Reserved	5 (6.1)	19 (16)	14 (16)
Positive profile	4 (6.1)	11 (16)	23 (16)

size than the first one and they describe networks with a small to average number of cliques and low transitivity, and networks with high transitivity and small number of cliques, respectively.

We found no differences between clusters regarding gender ($\chi^2 = 0.99$, $p = 0.60$, $df = 2$). With respect to age, there is a tendency for individuals having a network with low transitivity and an average number of cliques to be younger (37.4 years old), while those with high transitivity and small number of cliques are slightly older, with the smallest cluster of individuals occupying networks with low transitivity and high number of cliques is in the middle (40.3 and 38.1 years old, respectively). On a descriptive level, the participants with high transitivity and a small number of cliques expressed the highest Psychological Sense of Community, while the participants with low transitivity and a high number of cliques reported the lowest Psychological Sense of Community.

7.5 Types of personal networks and personality profiles

With the same classification procedure of the previous section,² we obtained three personality profiles after applying a cluster analysis (Table 5). The first cluster, consisting of 38% of the respondents have a positive profile, obtaining the comparatively highest score in each of the five personality factors. A second group, which also corresponds to 38% of the total, is characterized by having comparatively lower scores in extraversion and openness to experience, forming a profile of more reserved individuals. Finally, the smallest group, 14%, which stands out for having comparatively higher scores in neuroticism and lower scores in agreeableness, that is, more emotionally “unstable”³ individuals.

The comparison of individual variables according to the three profiles only yielded significant results in the case of degree centralization ($F_{2,97} = 5.195$, $p < 0.01$) and the Membership factor of the PSC scale ($F_{2,97} = 5.777$, $p < 0.01$). Specifically, from the unstable profile (cluster 1) to the positive profile (cluster 3), personal networks are successively less centralized and with higher scores in sense of belonging.

Finally, we looked at the relationship between clusters of networks and clusters of personality profiles. The results are presented in Table 6.

The result of chi-square test suggests that occurrences are statistically different from what would be expected by chance ($\chi^2 = 11.90$, $p = 0.01$, $df = 4$). Individuals with unstable profile

are overrepresented in the “low transitivity/high number of cliques” cluster and also in the “low transitivity/average number of cliques” cluster, albeit to a smaller degree. On the other hand, they are underrepresented in the cluster with “high transitivity/ small number of cliques.” The opposite pattern holds for participants with positive profile. The individuals belonging to the reserved personality cluster show a smaller discrepancy between empirical and expected frequencies in general, although there are slightly fewer in the “high transitivity/small number of cliques” cluster and slightly more in the “low transitivity/average number of cliques” cluster.

7.6 Triads and personality types

In this final analysis, we aim to explore at a descriptive level the relationship between different personality clusters and KR triads. The density measure is included, as it provides the information of the proportion of closed triads that contain ego. As shown in Figure 3, the clearest differences between clusters exist for purely structural measures (density, open and closed triads, all triads), implying that additional information about the weight of ties among alters is less informative for differentiation between personality clusters.

It is fitting to try to relate the proportion of closed triads with ego (density) in different personality clusters to Simmel's theory of “individuality” in dyads and triads (Wolff 1950, p. 137). According to Simmel, the dyad favors a relatively greater individuality of the members, allowing for greater “individualization.” Dyads “preserve the individuality” (Krackhardt, 1999 p. 185) in the sense that “no majority can outvote” one party, while an actor belonging to a triad is less free, less independent, than when she belongs to a dyad. Extrapolating from those characteristics of relational settings, Simmel makes an important distinction between two types of individualities regarding the setting they prefer. A “decided” (or qualitative) individuality refers to “singularity” and describes a person who will avoid groups where a majority may appear and will prefer multiple dyads. On the other hand, actors with a “strong” individuality will be more likely to look—given the choice—for a triadic than a dyadic setting. This theory could be easily connected with research on personality and networks. Our results suggest the Simmel's theory was right, if we relate “strong” personality with positive profile and “decided” personality with other two clusters, reserved and unstable. From that we can extrapolate specific prediction regarding preference of positive personality profile for closed, and other personality types for open triads.

Our results indicate that even when ego is excluded, the proportion of all triads (meaning closed and open triads taken together) and closed triads alone is higher in ego-networks of individuals with positive personality profile than for the other two personality clusters. Somewhat bigger differentiation of clusters is achieved with KR variant III in comparison with other two variants, which beside density controls for individual tendency for assigning weak and strong ties among alters. The networks of individuals with positive profile have more weak triads with one strong tie (WWS), and less open weak triads and mixed triads, than the other two personality clusters. The most likely explanation of a higher average proportion of WWS triads in this cluster is that the strong tie in a WWS triad constitutes a dyad with which ego is strongly connected. That is, a WWS triad containing three alters i , j and k , of which j and k are strongly connected is a proxy of a SSS triad in which ego is included (triad of ego, j , and k). Another general pattern visible in Figure 3 is that overall KR measures differentiate between the positive cluster and the other two clusters, reserved and unstable. However, the latter two seem to be almost undistinguishable by KR measures. Figure 3 shows that networks of individuals belonging to the reserved cluster have more of strong closed triads, when controlling for density and strength of ties (KR measure III), than individuals in the other two clusters. This finding may be explained by a possible tendency of reserved people to form their networks around connected actors (friends of friends), or around very close individuals, such as family members.

8. General discussion

In this research, we have observed a significant association between individual personality differences and the structural properties of personal networks. On the one hand, the highest scores on the five aggregate personality factors are inversely related to the degree centralization of personal networks. On the other hand, emotional stability is positively associated with density. All these findings show that individual personality differences are reflected in the cohesion of personal networks. This was confirmed with the building of typologies, which showed that individuals with a personality profile with a greater presence of neurotic features were less likely to have articulated personal networks (that is to say, organized in delineated clusters but with comparatively high levels of density). It is interesting to notice that of all the three personality clusters, people with reserved profile seem to be presented in each network cluster most similarly to what would be expected by chance. This pattern may be a relevant finding since it suggests the possibility that for this type of personality—less extraverted and less open to experience—psychological attributes are less associated with their networks. It could be because they are less spontaneous in the process of creation and maintenance of their networks, and/or because they simply “inherit” the networks of their alters. That, in turn, may lead to their networks being less a reflection of their psychological attributes. On a more general level, the group of people with reserved type of personality (38%) may be one of the reasons why the detection of a relationship between psychological attribute and social networks usually yields small effects and is often inconsistent. It is possible that there is no relationship between the psychological attributes and social networks for people with a reserved personality profile, while for the other two groups the association may exist and be higher than when looking at all of them together.

Regarding the introduction of other variables, besides the Big Five personality traits, in the exploration of the psychological attribute–social networks relationship, the important empirical observations of this study refer to Psychological Sense of Community. We verified that the density of the personal network is associated in a consistent way with the feeling of belonging to the neighborhood and, more broadly, with Psychological Sense of Community.

To summarize, when looking at the psychological dimensions of an individual and dimensions of individual's social network in the manner where different aspects are analyzed separately, we may “fail to see the bigger picture.” This was already recognized in both, network research literature on immigrants (Lubbers et al., 2007; Maya-Jariego, 2003), and the personality research. In the latter, the typological approach has been often contrasted with the traditional dimensional approach, but recently it has undergone a renaissance and the most prominent researchers (Costa et al., 2002) acknowledge that it may prove useful for labelling trait combinations that are associated with consequential outcomes. Those two trends taken together with our findings imply that typological approach to both psychological attributes and social networks may not only be more intuitive but also could be more useful and maybe even necessary when describing and understanding the interplay between individual psychological attributes and network structure.

In the analysis of triads, we used three different modifications of KR triadic census measure applied on alter–alter ties, suitable for networks with a fixed size. The three variants showed different patterns of association regarding the two groups of the measured psychological attributes. The important take-away message is that researchers need to be aware of which exact procedure they use when measuring triads—or indeed, any network property, as they may be differently related with individual psychological attributes.

Our findings extend those found by Kalish & Robins, (2006) that strong and closed triads in which ego is embedded are positively related with ego's emotional stability to the same triadic configuration between ego's alters (when using the first KR variant). However, we did not observe the association between both Emotional stability and Extraversion with WWW, which was found by Kalish & Robins for WWW with ego included. Instead, in our study, weak closed triads (WWW) of alters were related with Sense of Community and Conscientiousness, noting that the latter may be an indication of possible methodological artefacts.

Regarding open triads, we did not expect to arrive at a similar pattern of results as by Kalish & Robins, since an open triad with ego, in which ego is the “bridge,” is psychologically and socially different from an open triad between three ego’s alters. While Kalish & Robins reported a significant relation of open triads (where ego is a broker) with psychosocial traits (individual focus, locus of control), but not with the Big Five personality traits, we found that open triads among alters are associated with ego’s emotional stability and the mean score of the TIPI scale. SWN triads in the second KR variant and the mean score of the TIPI scale had the highest correlation found and negative, implying that individuals with high scores on all Big Five traits (“positive profile”) are less likely to report all three kinds of ties (absent tie, weak, and strong) appearing in the same triad of alters.

The most sophisticated measure of KR is the third variant which is based on individually tailored null model, and a generation of sample of 100 random networks in comparison to which the z-score for every triad is calculated. By using this variant, we controlled for density and the composition of ties strength, and uncovered new associations between specific triads and traits Agreeableness and Conscientiousness. Additionally, we constructed the measure of Non-randomness of the alter–alter ties in ego-network based on absolute z-scores of triadic motifs, and found it to be negatively associated with Agreeableness. A possible interpretation could be that people with higher Agreeableness tend to either inhabit, perceive or report more random triadic structures among their alters. Finally, descriptive analysis revealed that this KR variant differentiates between personality types more efficiently than the other two variants.

Although in our case we did not include ego in the analysis, the triad census showed a degree of relationship with psychological attributes similar to that originally observed by Kalish & Robins, (2006) and Staiano et al., (2012). We moved “one step” away from the ego—due to excluding ego’s ties to alters from the analysis—to its immediate social environment. Nevertheless, we found that an association between psychological attributes and differently measured triads of alters still exists. Therefore, we may interpret this finding as an additional “lead” that individual psychological attributes and social networks influence each other seeing that even ties in which ego is not directly involved are related with ego’s attributes.

It is important to note that these findings do not necessarily apply to other relationships in ego’s network, e.g. ties among ego’s alters and their friends (alters of alters) who are not directly connected to ego (two links away from ego). Thus, we are not able to generalize to all the triads in ego’s social life—but only to triads among his/her alters. Given that we did not measure ego’s ties in terms of strength, the direct comparison of different kinds of triads regarding their distance from ego is not possible and should be examined in future research.

8.1 Limitations of the study and future research

On the one hand, in our analysis, we implicitly regarded all alters as equally important. However, this is not the case and it may be that the proportions would be different if we looked at alters with only high importance (with strong tie to ego). There is no way to know where these triads are concentrated and among which alters exactly. Nonetheless, obtaining a fixed number of 45 alters seems an efficient strategy, which simultaneously allows for the examination of the most significant relationships for the individual, and at the same time obtaining a representation of the diversity of structures in personal networks. Another limitation of our study, typical of this kind of research, is a small sample. It can affect the range of observed correlations, and introduce a bias in the statistical inference (due to multi-testing). Although small, our sample is made of adults for whom we can assume that on a group level the psychological attributes, as well as networks, are more stable than it is the case of student population on which the data has been collected for some of the most notable previous studies.

We took advantage of well-established measures of network motifs often used in the application of network science in biology (Milo et al., 2002), to measure the network’s on meso level—the

level which is more appropriate for interpretation and theory development for our topic than the macro level captured by most network measures. These fine-grained and individually tuned measures of configurations may be more elegant and straightforward alternative to specifying the same exponential random graph model for each personal network and looking for meaningful differences in parameters between networks. It does not mean that we exhausted all possibilities of motif analysis. For example, including additional constraints (e.g. controlling for degree distribution in the ego-net) may yield to different observations about the relationship with psychological attributes. Therefore, future studies should further extend our approach of measuring different motifs in individual ego-network.

Our results are based on self-reported measures of one's network, making the network measure inherently subjective. From a psychological perspective, this subjectivity is not necessarily problematic as individual behavior is in large part determined by our own subjective experience. The reported networks are not, however, likely to be the exact reflection of reality, and there is always a possibility that people with certain traits have a tendency to see their social environments with specific bias (e.g. highly emotionally unstable people may underestimate the strength of ties between their alters). To investigate this interesting research question on its own, future research should incorporate objectively collected measures of social interactions. Furthermore, longitudinal data would allow examining the co-evolution of networks and individuals. In the same vein, incorporating the intensity of the relationship of ego with each alter, as well as the information about the affective quality of ties represent some of the many possible future research avenues.

9. Conclusion

Both typologies and census of triads showed to be effective strategies for describing the structural properties of personal networks and exploring their relationship to individual psychological differences. In this study, we observed that closed triads are positively correlated with sense of community, but alters embedded in triads with weak ties also seem to be of importance. When it comes to personality traits, emotional stability was positively correlated with strong closed triads and with the overall indicator of density; while agreeableness was associated with having less systematic clustering in personal networks.

The study of individual psychological differences in the structure of social networks is relatively new. So far, mainly the relationship between personality traits and the size, composition and connectivity of networks has been explored. However, there are other individual characteristics that a priori could be more directly related to the properties of networks, such as attachment styles and communication competences, as well as the skills for the initiation, development and maintenance of relationships (Bouchard & Maya-Jariego, 2019). This is the case of Psychological Sense of Community, which has to do with the subjective experience of the local environments and the relational settings (Sarason, 1974). Overall, we found a more significant association of the structural properties of personal networks with the sense of community than with personality traits.

Two properties of personal networks showed an especially significant association with Psychological Sense of Community: transitivity and a greater relative presence of closed triads of weak ties. The first one is not a surprise, since transitivity is one of the basic processes in the explanation of the evolution of social networks (Holland & Leinhardt, 1976, 1977; Snijders, 2001; Stokman & Doreian, 1997) and in the integration of personal networks (Louch, 2000). However, the connection of Sense of Community with the triads of weak ties is suggestive. Regardless of the value that small cohesive groups that provide support may have, the community seems to rely on a network of indirect ties between individuals who, without necessarily developing a strong personal relationship, can experience a shared sense of belonging.

Acknowledgments. The second author, Srebrenka Letina, is supported by the European Research Council (ERC) under the European Union's Horizon 2020 Research and Innovation Programme (Grant Agreement no. 648693).

Conflict of interest. Authors have nothing to disclose.

Supplementary. For supplementary material for this article, please visit <http://doi.org/10.1017/nws.2019.15>.

Notes

1 Given that research done by Kalish & Robins, (2006) is seminal and often referred to in this paper, sometimes the acronym Kalish & Robins will be used.

2 *k*-means cluster analysis, with a maximum of 10 iterations, using updated means and a convergence criterion of 0.02.

3 Despite its negative connotation, the term “unstable” was originally used by McCrae & Costa, (1987).

References

- Asendorpf, J. B., & Wilpers, S. (1998). Personality effects on social relationships. *Journal of Personality and Social Psychology*, 74(6), 1531–1544.
- Baker, L. R., & McNulty, J. K. (2011). Self-compassion and relationship maintenance: the moderating roles of conscientiousness and gender. *Journal of Personality and Social Psychology*, 100(5), 853–873. doi: [10.1037/a0021884](https://doi.org/10.1037/a0021884).
- Battistoni, E., & Fronzetti Colladon, A. F. (2014). Personality correlates of key roles in informal advice networks. *Learning and Individual Differences*, 34, 63–69.
- Benet-Martinez, V., & John, O. P. (1998). Los Cinco Grandes across cultures and ethnic groups: Multitrait-multimethod analyses of the Big Five in Spanish and English. *Journal of Personality and Social Psychology*, 75(3), 729.
- Bidart, C., & Lavenu, D. (2005). Evolutions of personal networks and life events. *Social Networks*, 27(4), 359–376.
- Bidart, C., Degenne, A., & Grossetti, M. (2018). Personal networks typologies: A structural approach. *Social Networks*, 54, 1–11.
- Borgatti, S. P., Everett, M. G., & Freeman, L. C. (2002). *Ucinet 6 for windows: Software for social network analysis*. Harvard, MA: Analytic Technologies.
- Bouchard, P., & Maya-Jariego, I. (2019). Estilos de apego, redes personales y satisfacción con la carrera profesional en el personal directivo de Quebec (Canadá). *Journal of Work and Organizational Psychology*. Accepted for publication.
- Burt, R. S., Jannotta, J. E., & Mahoney, J. T. (1998). Personality correlates of structural holes. *Social Networks*, 20, 63–87.
- Cohen, S., Brissette, I., Skoner, D., & Doyle, W. (2000). Social integration and health: The case of the common cold. *Journal of Social Structure*, 1(3), 1–7.
- Costa, P. T., Herbst, J. H., McCrae, R. R., Samuels, J., & Ozer, D. J. (2002). The replicability and utility of three personality types. *European Journal of Personality*, 16, 73–87.
- Demir, M., & Weitekamp, L. A. (2007). I am so happy 'cause today I found my friend: Friendship and personality as predictors of happiness. *Journal of Happiness Studies*, 8(2), 181–211.
- Dickens, W. J., & Perlman, D. (1981). Friendship over the life-cycle. *Personal Relationships*, 2, 91–122.
- Digman, J. M. (1990). Personality structure: Emergence of the five-factor model. *Annual Review of Psychology*, 41(1), 417–440.
- Doeven-Eggens, L., De Fruyt, F., Hendriks, A. J., Bosker, R. J., & Van der Werf, M. P. (2008). Personality and personal network type. *Personality and Individual Differences*, 45(7), 689–693.
- Emery, C. (2012). Uncovering the role of emotional abilities in leadership emergence. A longitudinal analysis of leadership networks. *Social Networks*, 34(4), 429–437.
- Emery, C., Calvard, T. S., & Pierce, M. E. (2013). Leadership as an emergent group process: A social network study of personality and leadership. *Group Processes & Intergroup Relations*, 16(1), 28–45.
- Fang, R., Landis, B., Zhang, Z., Anderson, M.H., Shaw, M.D., & Kilduff, M. (2015). Integrating personality and social networks: A meta-analysis of personality, network position, and work outcomes in organizations. *Organization Science*, 26(4), 1243–1260.
- Feiler, D. C., & Kleinbaum, A. M. (2015). Popularity, similarity, and the network extraversion bias. *Psychological Science*, 26, 593–603.
- Ferrand, A. (1989). Connaissances passagères et vieux amis. Les durées de vie des relations interpersonnelles. *Revue Suisse de Sociologie*, 2, 431–439.
- Furukawa, T., Sarason, I. G., & Sarason, B. R. (1998). Social support and adjustment to a novel social environment. *International Journal of Social Psychiatry*, 44(1), 56–70.
- Giannella, E., & Fischer, C. S. (2016). An inductive typology of egocentric networks. *Social Networks*, 47, 15–23.
- Goldberg, L. R. (1993). The structure of phenotypic personality traits. *American Psychologist*, 48(1), 26.

- Gosling, S. D., Rentfrow, P. J., & Swann, Jr., W. B. (2003). A very brief measure of the Big Five personality domains. *Journal of Research in Personality*, 37, 504–528.
- Holland, P. W., & Leinhardt, S. (1976). Local structure in social networks. *Sociological Methodology*, 7, 1–45.
- Holland, P. W., & Leinhardt, S. (1977). A dynamic model for social networks. *Journal of Mathematical Sociology*, 5(1), 5–20.
- Jason, L. A., Stevens, E., & Ram, D. (2015). Development of a three-factor psychological sense of community scale. *Journal of Community Psychology*, 43(8), 973–985.
- Kalish, Y. (2008). Bridging in social networks: Who are the people in structural holes and why are they there? *Asian Journal of Social Psychology*, 11, 53–66.
- Kalish, Y., & Robins, G. (2006). Psychological predispositions and network structure: The relationship between individual predispositions, structural holes and network closure. *Social Networks*, 28(1), 56–84.
- Klein, K. J., Lim, B. C., Saltz, J. L., & Mayer, D. M. (2004). How do they get there? An examination of the antecedents of centrality in team networks. *Academy of Management Journal*, 47(6), 952–963.
- Krackhardt, D. (1999). The ties that torture: Simmelian tie analysis in organizations. In S. B. Andrews, & D. Knoke (Eds.), *Research in the sociology of organizations*, vol. 16: *Networks in and around organizations* (pp. 183–210). Stamford: JAI Press.
- Krause, J., James, R., & Croft, D. P. (2010). Personality in the context of social networks. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 365(1560), 4099–4106.
- Landis, B. (2016). Personality and social networks in organizations: A review and future directions. *Journal of Organizational Behavior*, 37, S107–S121.
- Liu, Y., & Ipe, M. (2010). How do they become nodes? Revisiting team member network centrality. *The Journal of Psychology*, 144(3), 243–258.
- Louch, H. (2000). Personal network integration: Transitivity and homophily in strong-tie relations. *Social Networks*, 22(1), 45–64.
- Lozares, C., Martí, J., Molina, J. L., & García-Macías, A. (2013). La cohesión-integración versus la fragmentación social desde un perspectiva relacional. *Metodología de Encuestas*, 15, 57–75.
- Lubbers, M. J., Molina, J. L., & McCarty, C. (2007). Personal networks and ethnic identifications: The case of migrants in Spain. *International Sociology*, 22(6), 721–741.
- Maya-Jariego, I. (2002). Tipos de redes personales de los inmigrantes y adaptación psicológica. *REDES. Revista Hispana para el Análisis de Redes Sociales*, 1(4). DOI: <https://doi.org/10.5565/rev/redes.30> Retrieved from <https://revistes.uab.es/redes/article/view/v1-n1-maya>
- Maya-Jariego, I. (2003). *A general typology of the personal networks of immigrants with less than 10 years living in Spain*. Paper presented at the XXIII Sunbelt Conference, Cancún, Mexico.
- Maya-Jariego, I. (2004). Sentido de comunidad y potenciación comunitaria. *Apuntes de Psicología*, 22(2), 187–211.
- Maya-Jariego, I. (2018). Why name generators with a fixed number of alters may be a pragmatic option for personal network analysis. *American Journal of Community Psychology*, 62(1–2), 233–238. doi: [10.1002/ajcp.12271](https://doi.org/10.1002/ajcp.12271)
- Maya-Jariego, I., & Holgado, D. (2015). Living in the metropolitan area. Correlation of interurban mobility with the structural cohesion of personal networks and the originaive sense of community. *Psychosocial Intervention*, 24(3), 185–190.
- McCarty, C. (2002). Structure in personal networks. *Journal of Social Structure*, 3, 1.
- McCrae, R. R., & Costa, P. T. (1987). Validation of the five-factor model of personality across instruments and observers. *Journal of Personality and Social Psychology*, 52(1), 81–90.
- McMillan, B., & Chavis, D. M. (1986). Sense of community: A definition and theory. *Journal of Community Psychology*, 14, 6–23.
- Milo, R., Shen-Orr, S., Itzkovitz, S., Kashtan, N., Chklovskii, D., & Alon, U. (2002). Network motifs: Simple building blocks of complex networks. *Science*, 298, 824–827.
- Molina, J. L., Maya-Jariego, I., & McCarty, C. (2014). Giving meaning to social networks: Methodology for conducting and analyzing interviews based on personal network visualizations. In S. Dominguez, & B. Hollstein (Eds.), *Mixed methods social networks research. Designs and applications* (pp. 305–335). Cambridge: Cambridge University Press.
- Morgan, D. L., Neal, M. B., & Carder, P. (1997). The stability of core and peripheral networks over time. *Social Networks*, 19(1), 9–25.
- Neal, Z. P., & Neal, J. W. (2017). Network analysis in community psychology: Looking back, looking forward. *American Journal of Community Psychology*, 60, 279–295. doi: [10.1002/ajcp.12158](https://doi.org/10.1002/ajcp.12158)
- Regts, G., & Molleman, E. (2016). The moderating influence of personality on individual outcomes of social networks. *Journal of Occupational and Organizational Psychology*, 89(3), 656–682.
- Renau, V., Oberst, U., Gosling, S. D., Rusiñol, J., & Lusa, A. C. (2013). Translation and validation of the ten-item-personality inventory into Spanish and Catalan. *Aloma. Revista de psicologia, ciències de l'educació i de l'esport Blanquerna*, 31(2), 85–97.
- Roberts, S. G., Wilson, R., Fedurek, P., & Dunbar, R. I. M. (2008). Individual differences and personal social network size and structure. *Personality and Individual Differences*, 44(4), 954–964.
- Russell, D. W., Booth, B., Reed, D., & Laughlin, P. R. (1997). Personality, social networks, and perceived social support among alcoholics: A structural equation analysis. *Journal of Personality*, 65, 649–692.

- Sarason, S. B. (1974). *The psychological sense of community: Prospects for a community psychology*. San Francisco: Jossey Bass.
- Selden, M., & Goodie, A. S. (2018). Review of the effects of Five Factor model personality traits on network structures and perceptions of structure. *Social Networks*, 52, 81–99.
- Selthout, M., Burk, W., Branje, S., Denissen, J., Van Aken, M., & Meeus, W. (2010). Emerging late adolescent friendship networks and Big Five personality traits: A social network approach. *Journal of Personality*, 78(2), 509–538.
- Siltaloppi, J., & Vargo S.L. (2017). Triads: A review and analytical framework. *Marketing Theory*, 17(4), 395–414.
- Simmel, G. (1950). *The sociology of Georg Simmel*. New York: Free Press.
- Snijders, T. A. B. (2001). The statistical evaluation of social network dynamics. *Sociological Methodology*, 31(1), 361–395.
- Staiano, J., Pianesi, F., Lepri, B., Sebe, N., Aharony, N., & Pentland, A. (2012). Friends don't lie. *Proceedings of the 2012 ACM Conference on Ubiquitous Computing – UbiComp'12*, Pittsburgh, PA, USA: ACM.
- Stokman, F. N., & Doreian, P. (1997). Evolution of social networks: Processes and principles. In P. Doreian, & F. N. Stokman (Eds.), *Evolution of social networks* (pp. 233–250). Amsterdam, The Netherlands: Gordon and Breach Publishers.
- Swickert, R. J., Rosentreter, C. J., Hittner, J. B., & Mushrush, J. E. (2002). Extraversion, social support processes, and stress. *Personality and Individual Differences*, 32(5), 877–891.
- Totterdell, P., Holman, D., & Hukin, A. (2008). Social networkers: Measuring and examining individual differences in propensity to connect with others. *Social Networks*, 30(4), 283–296.
- Tupes, E. C., & Christal, R. E. (1961). Recurrent personality factors based on trait ratings. USAF ASD Technical Report (No. 61-97), Lackland Airforce Base, TX: US Air Force.
- Tupes, E. C., & Christal, R. E. (1992). Recurrent personality factors based on trait ratings. *Journal of Personality*, 60(2), 225–251.
- van der Linden, D., te Nijenhuis, J., & Bakker, A. B. (2010). The general factor of personality: A meta-analysis of Big Five intercorrelations and a criterion-related validity study. *Journal of Research in Personality*, 44, 315–327.
- Vinacke, W. E., & Arkoff, A. (1957). An experimental study of coalitions in the triad. *American Sociological Review*, 22(4), 406–414.
- Wagner, J., Lüdtke, O., Roberts, B. W., & Trautwein, U. (2014). Who belongs to me? Social relationship and personality characteristics in the transition to young adulthood. *European Journal of Personality*, 28(6), 586–603.
- Wellman, B. (1983). Network analysis: Some basic principles. *Sociological Theory*, 1, 155–200.
- Wolff, K. H. (1950). *The sociology of Georg Simmel*. Glencoe, IL: The Free Press.
- Wrzus, C., Zimmermann, J., Mund, M., & Neyer, F. J. (2017). Friendships in young and middle adulthood: Normative patterns and personality differences. In M. Hojjat, & A. Moyer (Eds.), *Psychology of friendship* (pp. 21–38). New York: Oxford University Press.
- Xia, L., Yuan, Y. C., & Gay, G. (2009). Exploring negative group dynamics: Adversarial network, personality, and performance in project groups. *Management Communication Quarterly*, 23(1), 32–62.
- Zhu, X., Woo, S. E., Porter, C., & Brzezinski, M. (2013). Pathways to happiness: From personality to social networks and perceived support. *Social Networks*, 35(3), 382–393.